COS 433: Cryptography

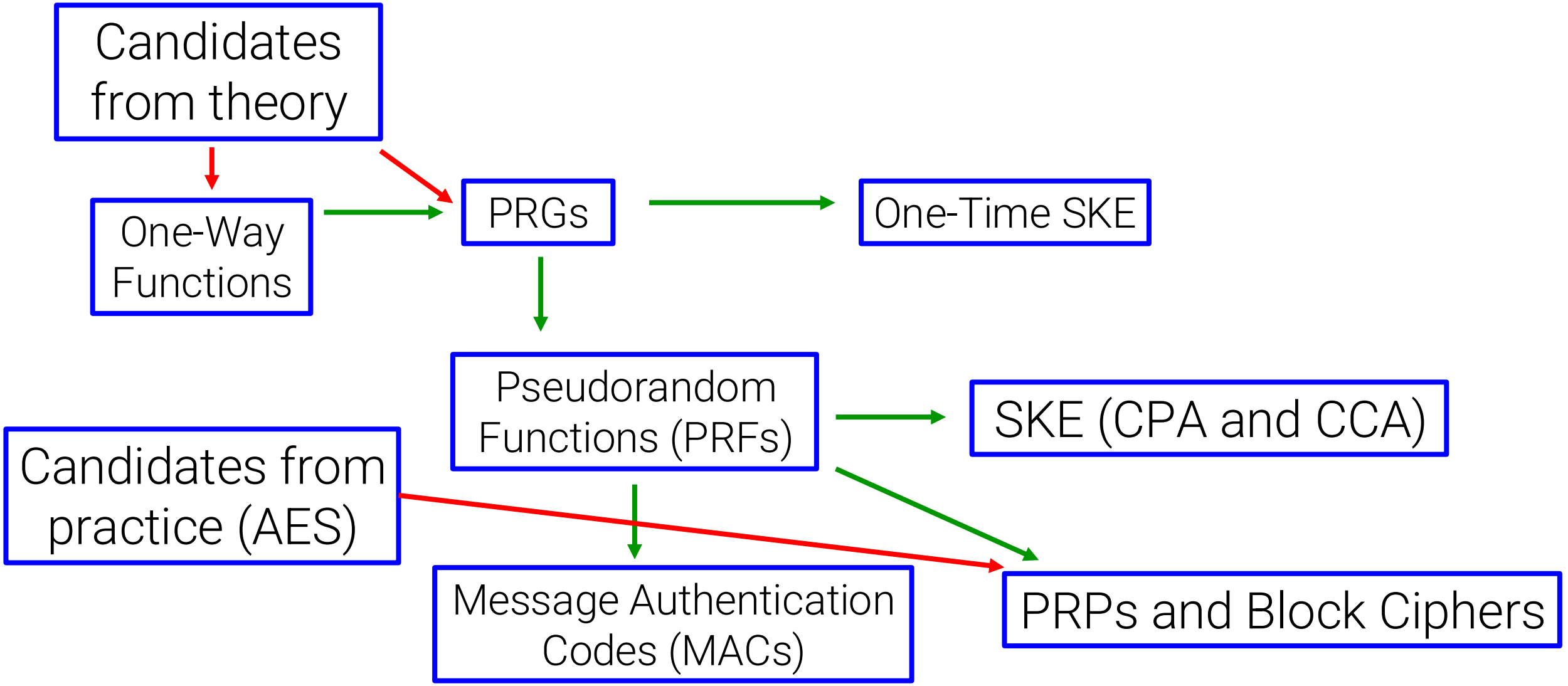
Announcements:

- Problem Set 3 due 10/28

- Lecture notes on computational number theory (by Dana Angluin) on the course website

Recap

Computationally Secure Cryptography



Next: Number-Theoretic Cryptography

Linear algebra over $\mathbb{Z}_2$

Security: dimension ($\mathbb{Z}_2^\lambda$)

Before

Arithmetic over $\mathbb{Z}_N$

Security: $N \approx 2^\lambda$

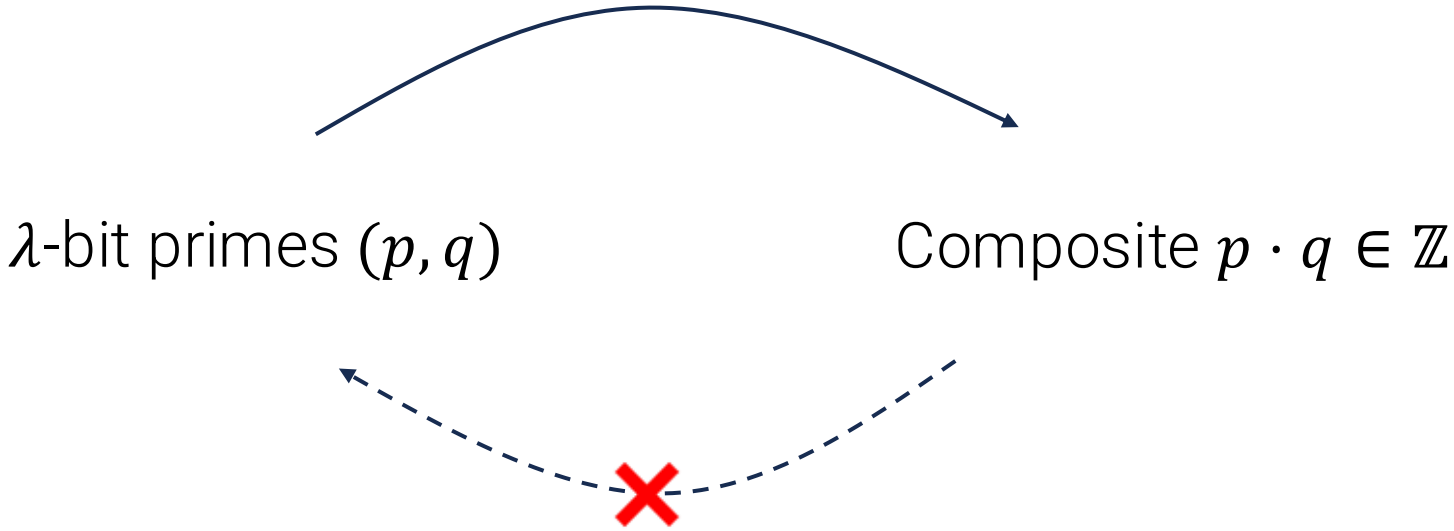
Now

Why Number-Theoretic Cryptography?

- Algorithmic number theory is over 2000 years old → good source of cryptographic candidates
- Direct constructions of more powerful primitives (SKE and PRFs)
- Used to build public-key cryptography (even in practice)

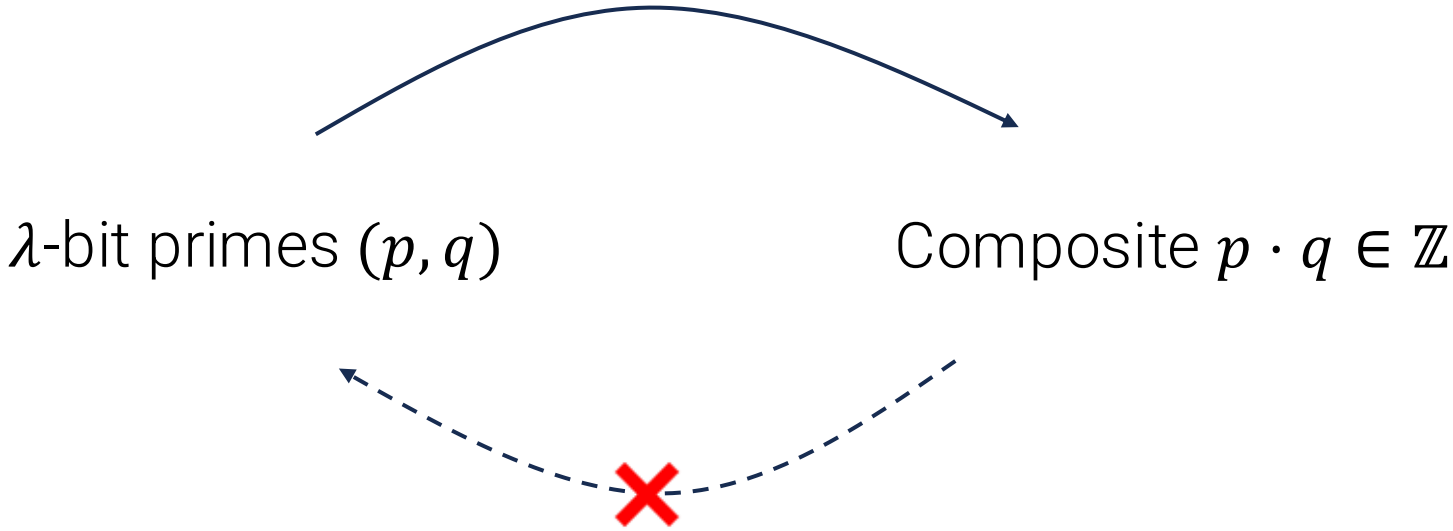
OWFs based on Factoring

OWFs based on Factoring



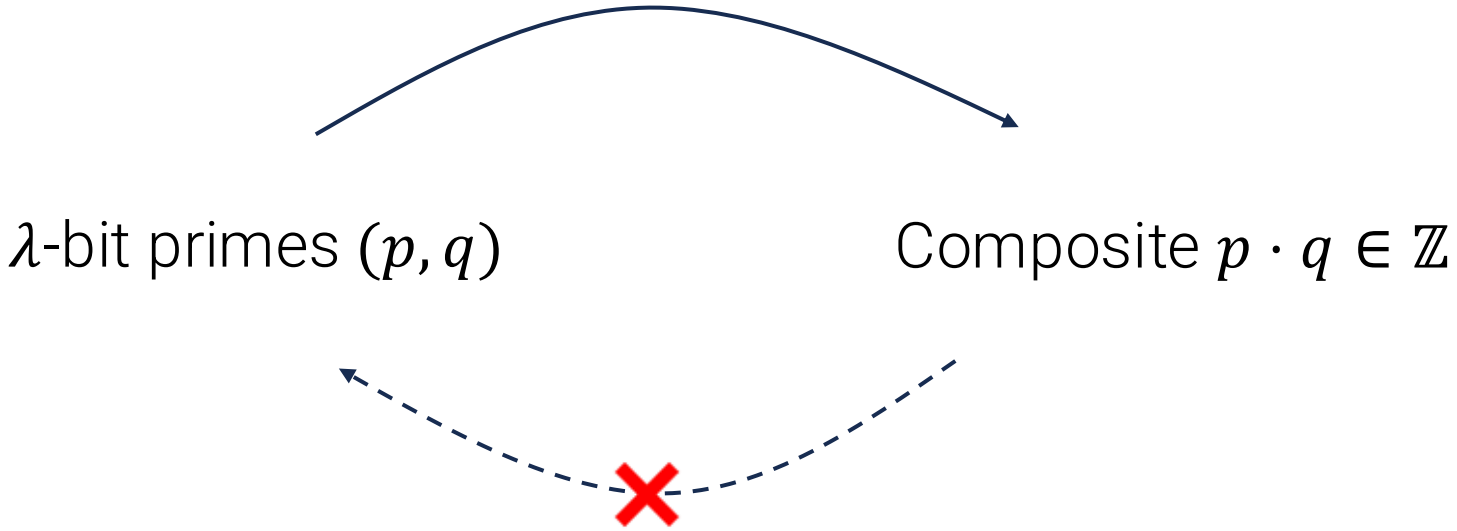
- Integer multiplication is easy ($O(\lambda^2)$)
- Prime factorization is believed to be hard (p, q random λ-bit primes)

OWFs based on Factoring



- How do we turn this into an actual OWF? (Need binary string domain)
- Fact 1: roughly $1/\lambda$ fraction of λ-bit integers are primes (Prime Number Theorem)
- Fact 2: primality testing is in P (Miller-Rabin, Agrawal-Kayal-Saxena)

OWFs based on Factoring



$f: \{0,1\}^{2 \cdot \lambda^3} \to \{0,1\}^{2 \cdot \lambda}$:

- Interpret input as $(p_1, p_2, \ldots, p_{\lambda^2}, q_1, q_2, \ldots, q_{\lambda^2})$
- For i from 1 to λ, check if p_i, q_i are prime.
  - Set p to be the first prime p_i, same for q
- Output $p \cdot q$

Key point: (p, q) extremely close to a uniformly random pair of primes

Plan for today

- Understand the hardness of factoring (a little) better
  - Crash course in computational number theory
- Find better number theoretic cryptographic constructions

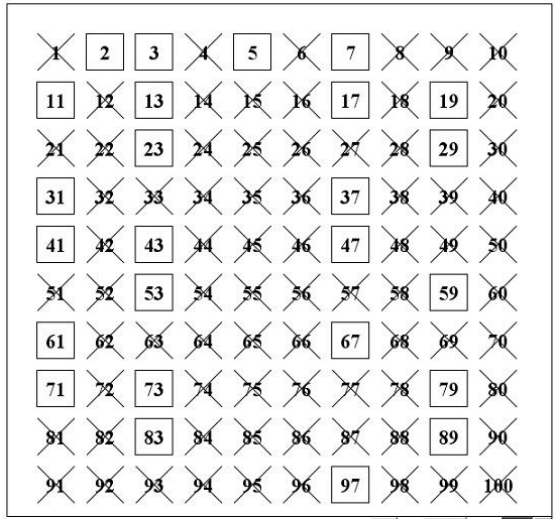
Algorithms for Factoring N (history)

1. Trial division

- For a from 2 to $N-1$, check if a divides N evenly
- Running time: $N \cdot \mathbf{poly}(\lambda) = 2^{\lambda} \cdot \mathbf{poly}(\lambda)$ for $N \leq 2^{\lambda}$
- Optimization: only check a from 2 to $\sqrt{N}$. Running time: $2^{\lambda/2} \cdot \mathbf{poly}(\lambda)$

2. Sieve of Eratosthenes

- Only divide N by primes
- Build a table of all primes up to $\sqrt{N}$
- Bonus: table is reusable
- More practical but asymptotically similar



Algorithms for Factoring N

3. Fermat's Method

- Find integers a, b such that $N = a^2 - b^2 = (a + b)(a - b)$
- Requires brute force up to $a, b = \sqrt{N}$ (slow)

4. Congruence of squares (best known approach)

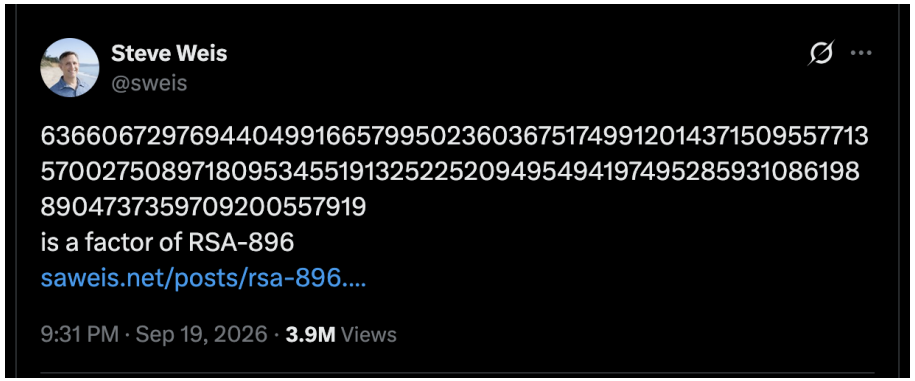
- Find a, b such that but $a^2 - b^2 \equiv 0 \pmod{N}$ but $a \not\equiv \pm b \pmod{N}$
- Then, $\gcd(a + b, N)$ is a proper divisor of N
  - Euclid's algorithm computes $\gcd$ in $\mathbf{poly}(\lambda)$ time
- How to find (a, b)? Complicated
  - Quadratic Sieve [Dixon '81, Pomerance '81] runs in time $2^{O\left(\sqrt{\lambda \cdot \log \lambda}\right)}$
  - Number Field Sieve [Pollard '88] runs heuristically in time $2^{O\left(\sqrt[3]{\lambda \cdot \log^2(\lambda)}\right)}$

Algorithms for Factoring N

Heuristic best known time complexity of factoring: $2^{O\left(\sqrt[3]{\lambda \cdot \log^2(\lambda)}\right)}$

- Current big factoring challenge: random $p, q \leq 2^{\lambda}$ with $\lambda = 512$ (RSA-1024)
  - New records from last month: $\lambda = 431$ (RSA-260), $\lambda = 448$ (RSA-896)





- "Safe" choices: $\lambda = 1024$ (until 2030), $\lambda = 2048$ (until 2038)

Hardness of Factoring

Evidence for factoring being computationally hard?

- Mathematicians have been been trying to factor for 2000+ years
- The world has been trying to factor big numbers since at least 1978
- Relate factoring to other problems in computational number theory

Theorem:

Factoring $N \Leftrightarrow$ Solving quadratic equations over $\mathbb{Z}_N$

Arithmetic over $\mathbb{Z}_N$

- $\mathbb{Z}_N$ is a commutative ring
  - Addition and multiplication operations, commutative and associative
  - Additive identity $\mathbf{0}$, additive inverses $a + (-a) = 0$
  - Multiplication distributes over addition
- $\mathbb{Z}_N$ is an effective family of rings (we can compute on $\mathbb{Z}_N$)
  - Elements have a canonical string representative
  - Addition, subtraction and multiplication take $\mathbf{poly}\log(N) = \mathbf{poly}(\lambda)$ time.

Arithmetic over $\mathbb{Z}_N$

How hard is it to solve equations in one variable over $\mathbb{Z}_N$?

- Case 1: linear equations

Given a, b, find x such that $a \cdot x \equiv b \pmod{N}$

- Solution exists iff $\gcd(a, N)$ divides b
- If $\gcd(a, N) = 1$ (wlog), there exists a^{-1} such that $a \cdot a^{-1} \equiv 1 \pmod{N}$
  - Extended Euclidean algorithm finds a^{-1} in $\mathrm{poly}(\lambda)$ time
  - $x = a^{-1} \cdot b$
- For later: define $\mathbb{Z}_N^\times = \{a \in \mathbb{Z}_N : \gcd(a, N) = 1\}$ (multiplicative group)

Arithmetic over $\mathbb{Z}_N$

How hard is it to solve equations in one variable over $\mathbb{Z}_N$?

- Case 2: quadratic equations

Given a, b, c, find x such that $ax^2 + bx + c \equiv 0 \pmod{N}$

Square root finding over $\mathbb{Z}_N$

How hard is it to solve equations in one variable over $\mathbb{Z}_N$?

- Case 2: quadratic equations (simplifies to:)

Given a, find x such that $x^2 \equiv a \pmod{N}$

Claim 1: easy if N is prime

Claim 2: easy if factoring N is easy

Claim 3: hard if factoring N is hard

Square root finding over $\mathbb{Z}_N$

How hard is it to solve equations in one variable over $\mathbb{Z}_N$?

- Case 2: quadratic equations (simplifies to:)

Given a, find x such that $x^2 \equiv a \pmod{N}$

Claim 1: easy if N is prime

Fermat's little theorem: $a^p \equiv a \pmod{p}$ for all a. $a^{p-1} \equiv 1$ for all nonzero a.

Proof: $(x+y)^p = \sum_{k=0}^{p} \binom{p}{k} x^k y^{p-k} \equiv x^p + y^p \pmod{p}$ by the binomial theorem.

Square root finding over $\mathbb{Z}_N$

How hard is it to solve equations in one variable over $\mathbb{Z}_N$?

- Case 2: quadratic equations (simplifies to:)

Given a, find x such that $x^2 \equiv a \pmod{N}$

Claim 1: easy if N is prime

Special case: if $N = p \equiv 3 \pmod{4}$, then

$$\left(a^{\frac{p+1}{4}}\right)^2 \equiv a^{\frac{p+1}{2}} \equiv a \cdot a^{\frac{p-1}{2}} \pmod{p}$$

Fact: a is a square mod p if and only if $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$

Square root finding over $\mathbb{Z}_N$

How hard is it to solve equations in one variable over $\mathbb{Z}_N$?

- Case 2: quadratic equations (simplifies to:)

Given a, find x such that $x^2 \equiv a \pmod{N}$

Claim 1: easy if N is prime

Special case: if $N = p \equiv 3 \pmod{4}$ and a is a square mod p, then

$$\left(a^{\frac{p+1}{4}}\right)^2 \equiv a \pmod{p}$$

General case: exercise

Exponentiation is easy by repeated squaring

Square root finding over $\mathbb{Z}_N$

How hard is it to solve equations in one variable over $\mathbb{Z}_N$?

- Case 2: quadratic equations (simplifies to:)

$$\text{Given } a, \text{ find } x \text{ such that } x^2 \equiv a \pmod{N}$$

Claim 2: easy if factoring N is easy

Proof: Chinese Remainder Theorem

If $\gcd(N, M) = 1$, then the map

$$x \mapsto (x \bmod N, x \bmod M)$$

Is an efficiently invertible bijection from $\mathbb{Z}_{NM} \to \mathbb{Z}_N \times \mathbb{Z}_M$

Square root finding over $\mathbb{Z}_N$

How hard is it to solve equations in one variable over $\mathbb{Z}_N$?

- Case 2: quadratic equations (simplifies to:)

Given a, find x such that $x^2 \equiv a \pmod{N}$

Claim 2: easy if factoring N is easy

Proof: special case $N = pq$

Solution exists iff a is a square mod p and mod q

Find solutions mod p and mod q (up to 2 each)

Piece together (up to 4) solutions mod N by inverting the CRT map

Square root finding over $\mathbb{Z}_N$

How hard is it to solve equations in one variable over $\mathbb{Z}_N$?

- Case 2: quadratic equations (simplifies to:)

$$\text{Given } a, \text{ find } x \text{ such that } x^2 \equiv a \pmod{N}$$

Key point: solutions (for $N = pq$) have the form

$$x \equiv \pm r \pmod{p}, x \equiv \pm s \pmod{q}$$

Square root finding over $\mathbb{Z}_N$

Given a, find x such that $x^2 \equiv a \pmod{N}$

Claim 3: hard (even on average) if factoring N is hard

Proof: by reduction. Suppose that A is an efficient algorithm with

$$\Pr_{\substack{p,q \\ x \leftarrow \mathbb{Z}_N^\times}} [A(pq, x^2) = y \text{ such that } y^2 \equiv x^2 \pmod{N}] = \epsilon(\lambda)$$

Goal: build A' that factors N with non-negligible probability

Define:

$$A'(N) = \gcd\big(N, x + A(N, x^2)\big)$$

Square root finding over $\mathbb{Z}_N$

Goal: build A' that factors N with non-negligible probability

Define:

$$A'(N) = \gcd\left(N, x + A(N, x^2)\right)$$

Analysis: let $y = A(N, x^2)$.

1) We know that $y^2 \equiv x^2$ with probability ϵ.

2) There are four possible solutions for a fixed $z = x^2$

3) Subroutine A does not know, given z, which of the four x that A' used.

Square root finding over $\mathbb{Z}_N$

Goal: build A' that factors N with non-negligible probability

Define:

$$A'(N) = \gcd\left(N, x + A(N, x^2)\right)$$

Analysis: let $y = A(N, x^2)$.

1) We know that $y^2 \equiv x^2$ with probability ϵ.

$$\Pr[\, y \equiv x \pmod N \mid y^2 \equiv x^2] = 1/4$$

$$\Pr[\, y \equiv -x \pmod N \mid y^2 \equiv x^2] = 1/4$$

Square root finding over $\mathbb{Z}_N$

Goal: build A' that factors N with non-negligible probability

Define:

$$A'(N) = \gcd\left(N, x + A(N, x^2)\right)$$

Analysis: let $y = A(N, x^2)$.

1) We know that $y^2 \equiv x^2$ with probability ϵ.

$$\Pr[\, y \notin \{\pm x \pmod N\} \mid y^2 \equiv x^2] = 1/2$$

2) But for any solution y, we have $y \equiv \pm x \pmod p$ and $\pmod q$

3) For these y, $y + x$ is divisible by exactly one of p and q!

Theorem:

Factoring $N \Leftrightarrow$ Solving quadratic equations over $\mathbb{Z}_N$

Back to constructing cryptography

OWFs based on Factoring

We believe this (modulo domain issues) is a OWF:

$$f(p, q) = p \cdot q$$

OWFs based on Factoring

Here's another candidate OWF:

$$f(p, q, x) = (pq, x^2 \bmod pq)$$

More secure: can't even find a square root x, let alone (p, q)

Question: is this a "PRG?"

$$f(p, q, x) \approx_c (pq, u \bmod pq)?$$

Pseudorandomness based on Factoring?

Question: is it the case that

$$(pq, x^2 \bmod pq) \approx_c (pq, u \bmod pq)?$$

Concern: $\gcd(u, pq) > 1$ can be efficiently tested

- This is ok: the probability of this event is negligible.

Answer: no!

- u is a square modulo p, q with probability ½ (independently)
- Efficient algorithm calculating parity of these two bits of info

Pseudorandomness based on Factoring?

Definition: the Jacobi symbol

For $N = p_1 \cdot \ldots \cdot p_k$, and $\left(\frac{a}{N}\right) = \left(\frac{a}{p_1}\right) \cdot \ldots \cdot \left(\frac{a}{p_k}\right)$

$\left(\frac{a}{p}\right) = \pm 1$ depending on whether a is a square mod p

Fact: if a is a square mod N, then $\left(\frac{a}{N}\right) = 1$.

Fact: $\mathbf{E}_{u \leftarrow \mathbb{Z}_N^\times} \left(\frac{u}{N}\right) = 0$

Pseudorandomness based on Factoring

Definition: the Jacobi symbol

For $N = p_1 \cdot \ldots \cdot p_k$, and $\left(\frac{a}{N}\right) = \left(\frac{a}{p_1}\right) \cdot \ldots \cdot \left(\frac{a}{p_k}\right)$

$\left(\frac{a}{p}\right) = \pm 1$ depending on whether a is a square mod p

Crazy Fact: $\left(\frac{a}{N}\right)$ is efficiently computable given (a, N)

Reason: for odd (a, N): $\left(\frac{a}{N}\right) = (-1)^{\frac{a-1}{2} \cdot \frac{N-1}{2}} \cdot \left(\frac{N}{a}\right)$ (quadratic reciprocity)

Pseudorandomness based on Factoring?

Question: is it the case that

$$(pq, x^2 \bmod pq) \approx_c (pq, u \bmod pq)?$$

Answer: no!

But this is the only known “obstacle” to pseudorandomness.

Pseudorandomness based on Factoring?

Conjecture (Quadratic Residuosity):

$$(pq, x^2 \bmod pq) \approx_c (pq, u \bmod pq)$$

Where u is a uniform element of $\mathbb{Z}_N^\times$ subject to the condition that

$$\left(\frac{u}{N}\right) = 1$$

Pseudorandomness based on Factoring?

Assumption (Quadratic Residuosity):

$$(pq, x^2 \bmod pq) \approx_c (pq, u \bmod pq)$$

Where u is a uniform element of $\mathbb{Z}_N^\times$ subject to the condition that

$$\left(\frac{u}{N}\right) = 1$$

Pseudorandomness based on Factoring?

Assumption (Quadratic Residuosity, equivalent formulation):

$$(pq, x^2 \bmod pq) \approx_c (pq, a \cdot x^2 \bmod pq)$$

Where a is any fixed element of $\mathbb{Z}_N^\times$ such that $\left(\frac{a}{N}\right) = 1$

Equivalent because:

- if a is a square, $a \cdot x^2$ is a random square.
- if a is not a square, $a \cdot x^2$ is uniform non-square with Jacobi symbol 1

Pseudorandomness based on Factoring?

Assumption (Quadratic Residuosity, equivalent formulation):

$$(pq, x^2 \bmod pq) \approx_c (pq, a \cdot x^2 \bmod pq)$$

Where a is any fixed element of $\mathbb{Z}_N^\times$ such that $\left(\frac{a}{N}\right) = 1$

Example:

- if $p, q \equiv 3 \pmod 4$, $a = -1$ is a non-square modulo p and q

Pseudorandomness based on Factoring?

Assumption (Quadratic Residuosity, equivalent formulation):

$$(pq, x^2 \bmod pq) \approx_c (pq, -x^2 \bmod pq)$$

Where a is any fixed element of $\mathbb{Z}_N^\times$ such that $\left(\frac{a}{N}\right) = 1$

Example:

- if $p, q \equiv 3 \pmod 4$, $a = -1$ is a non-square modulo p and q
- note: if factoring is hard, then it is still hard subject to $p, q \equiv 3 \pmod 4$

Pseudorandomness based on Factoring?

Question: Are the following equivalent?

Factoring assumption $\Leftrightarrow$ QR assumption

Open Problem

Quadratic Residuosity

Lemma (QR worst-case to average-case):

Suppose that $A(N, z)$ violates the QR assumption for a fixed N.
Then, there exists an efficient A' (independent of N) such that:

$A'(N, z)$ decides whether **any given** z is a square modulo N

Proof:

- Given z, compute $\left(\frac{z}{N}\right)$. Output "non-square" if -1.
- Otherwise, do the following $\mathbf{poly}(\lambda)$ times:
  - Run $A(N, z \cdot x^2)$ for a uniformly random x.
- Output the majority answer of the calls to A.

Quadratic Residuosity

Assumption (Quadratic Residuosity, equivalent formulation):

$$(pq, x^2 \bmod pq) \approx_c (pq, a \cdot x^2 \bmod pq)$$

Where a is any fixed element of $\mathbb{Z}_N^\times$ such that $\left(\frac{a}{N}\right) = 1$

Not exactly a PRG, but we can use this to encrypt messages!

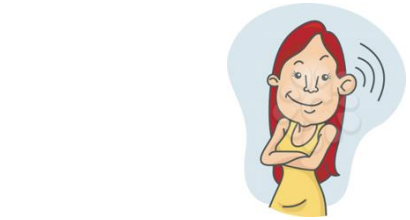
Encryption from QR (Goldwasser-Micali)

Encryption from QR



$k = (p, q)$

$m \in \{+1\}$





$k = (p, q)$

- Key: rand $N = pq$ is a “Blum integer”



Encryption from QR



$k = (p, q)$
$m \in \{\pm 1\}$

$$\mathrm{ct} = r^2 \cdot m \pmod{N}$$





$k = (p, q)$

$$m' = \left(\frac{\mathrm{ct}}{p}\right)$$

- Key: random primes p, q with $p \equiv q \equiv 3 \pmod 4$. $N = pq$ is a "Blum integer"
- To encrypt a one-bit message $m \in \{\pm 1\}$:
  - Sample a random $r \leftarrow \mathbb{Z}_N^\times$.
  - Output $r^2 \cdot m \bmod N$
- To decrypt, check if ct is a square modulo N (equivalently, modulo p or q)

Encryption from QR



$k = (p, q)$
$m \in \{\pm 1\}$

$$\mathrm{ct} = r^2 \cdot m \pmod N$$





$k = (p, q)$

$$m' = \left(\frac{\mathrm{ct}}{p}\right)$$

- Claim: this scheme is CPA-secure!
- One-message security: $r^2 \cdot m \bmod N \approx_c r^2 \cdot m' \bmod N$ by QR.
- Many-message IND-security:
  - Want: $(r_1^2 \cdot m_1 \bmod N, \ldots, r_t^2 \cdot m_t) \bmod N \approx_c (r_1^2 \cdot m_1', \ldots, r_t^2 \cdot m_t') \bmod N$
  - In fact, we claim: $(N, r_1^2 \cdot m_1, \ldots, r_t^2 \cdot m_t) \approx_c (N, r_1^2 \cdot m_1', \ldots, r_t^2 \cdot m_t')$
  - Hybrid argument: change m_i one at a time using QR.

Encryption from QR



$k = (p, q)$
$m \in \{\pm 1\}$

$$\mathrm{ct} = r^2 \cdot m \pmod{N}$$





$k = (p, q)$

$$m' = \left(\frac{\mathrm{ct}}{p}\right)$$

This was really easy! (compared to building/using PRFs)

Why?

Encryption from QR



$k = (p, q)$
$m \in \{\pm 1\}$

$$\mathrm{ct} = r^2 \cdot m \pmod{N}$$





$k = (p, q)$

$$m' = \left(\frac{\mathrm{ct}}{p}\right)$$

Observation 1: Alice only needs N to be able to encrypt! (Not p, q)

Encryption from QR



$N = pq$
$m \in \{\pm 1\}$

$$\mathrm{ct} = r^2 \cdot m \pmod{N}$$





$k = (p, q)$

$$m' = \left(\frac{\mathrm{ct}}{p}\right)$$

Observation 1: Alice only needs N to be able to encrypt! (Not p, q)

Observation 2: secure even if the adversary knows N! (but not p, q)

Encryption from QR



$N = pq$
$m \in \{\pm 1\}$

$$\mathrm{ct} = r^2 \cdot m \pmod{N}$$



$N = pq$



$k = (p, q)$

$$m' = \left(\frac{\mathrm{ct}}{p}\right)$$

Observation 1: Alice only needs N to be able to encrypt! (Not p, q)

Observation 2: secure even if the adversary knows N! (but not p, q)

Encryption from QR



$N = pq$
$m \in \{\pm 1\}$

$$\mathrm{ct} = r^2 \cdot m \pmod{N}$$



$N = pq$



$k = (p, q)$

$$m' = \left(\frac{\mathrm{ct}}{p}\right)$$

In other words: this scheme can have a Public Encryption Key

Encryption from QR





$m \in \{\pm 1\}$

$\mathrm{pk} = N = p \cdot q$ (broadcast)

$\mathrm{ct} = r^2 \cdot m \pmod{N}$



$\mathrm{sk} = (p, q)$

$m' = \left(\frac{\mathrm{ct}}{p}\right)$

In other words: this scheme can have a Public Encryption Key

Encryption from QR





$m \in \{\pm 1\}$

$\mathrm{pk} = N = p \cdot q$ (broadcast)

$\mathrm{ct} = r^2 \cdot m \pmod{N}$



$\mathrm{sk} = (p, q)$

$m' = \left(\frac{\mathrm{ct}}{p}\right)$

No need to share a key in advance!

Continue next time